

Started on Monday, 29 November 2021, 1:11 PM

State Finished

Completed on Monday, 29 November 2021, 1:11 PM

Time taken 19 secs

Marks 0/15

Grade 0 out of 100

Question **1**

Not answered

Marked out of 1

In secondary research studies, which of the following items is the unit of recruitment?

Select one:

- ☐ Clusters distributed geographically
- ☐ Secondary health care centres
- ☐ Individual patients
- ☐ Institutions
- ☐ Previously completed studies

Your answer is incorrect.

Secondary research refers to research conducted using data from other researches. Here individual patients are not recruited, instead eligible studies are 'recruited'.

The correct answer is: Previously completed studies

Question 2

Not answered

Marked out of 1

An old age psychiatrist is interested in reviewing the literature on hyponatraemia related to SSRI use. He has wide clinical experience on this topic and chooses some key review papers as starting point and collates the information known to him already to provide a summary of evidence. Which research method is he following?

Select one:

- ☐ Meta-analysis
- ☐ Qualitative research
- ☐ Case series
- ☐ Systematic review
- ☐ Narrative review

Your answer is incorrect.

Narrative reviews are usually carried out by experts in the field of study and are more or less close to an expert's opinion supported by research evidence. These reviews are generally broad without a narrow or specific clinical question and are subjective. They carry a higher risk of being prone to bias than systematic reviews. Nevertheless narrative reviews can provide a good introduction to a topic, stimulate interest and controversies and can even collate existing data for generating new hypotheses.

The correct answer is: Narrative review

Question 3

Not answered

Marked out of 1

Which of the following is a characteristic feature of narrative review but not systematic review?

Select one:

- ☐ Quantitative synthesis of data
- ☐ Uniform criteria of study selection
- ☐ Study a focussed narrow question
- ☐ Comprehensive and specific data collection
- ☐ Literature search based on expert's familiarity

Your answer is incorrect.

Narrative review usually involves a patchy literature search. Literature is not searched methodically but rather in a random manner according to one's familiarity and to suit one's flow of argument. Because of the above, narrative reviews cannot be replicated (unless repeated by the same author!!).

The correct answer is: Literature search based on expert's familiarity

Question **4**

Not answered

Marked out of 1

The study design that is most frequently combined in meta-analysis is

Select one:

- ☐ Cohort studies
- ☐ Case control studies
- ☐ RCTs
- ☐ Case series
- ☐ Uncontrolled trials

Your answer is incorrect.

It is usually RCTs that undergo meta-analysis. If the primary studies included have poor quality, then even if sound methodological and statistical procedures are followed the outcome may be meaningless. This is called GIGO principle of 'garbage in, garbage out'!!

The correct answer is: RCTs

Question 5

Not answered

Marked out of 1

Which of the following factor is the least/not important in determining the quality of a meta-analysis?

Select one:

- ☐ Sample size of individual studies
- ☐ Comparability of intervention among individual studies
- ☐ Magnitude of outcome reported from individual studies
- ☐ Methodological quality of individual studies
- ☐ Total number of individual studies

Your answer is incorrect.

While determining the quality of a meta-analysis, one can consider 1. The nature of the patient sample, 2. Outcomes studied, 3. Lengths of follow-up and 4. Comparability (e.g. dosage) of treatment. Methodological factors include adequacy of sample randomisation, adequate concealment and control of intervention, analysis with intention to treat principles, and having an objective, blinded, outcome assessment. The magnitude of outcomes as such does not reflect the quality of trials.

The correct answer is: Magnitude of outcome reported from individual studies

Question 6

Not answered

Marked out of 1

Weighting refers to the level of significance attached to the outcome of individual studies considered in a meta-analysis. Which of the following factors are considered for weighting?

Select one:

- ☐ All of the listed options
- ☐ Impact factor of the published journal
- ☐ Sample size of studies
- ☐ Methodological rigour of studies
- ☐ Precision of results

Your answer is incorrect.

Weighting refers to the significance attached to each study based on sample size, precision, external validity (the extent to which results are generalisable) and methodological quality. It is assigned independent the of outcome of a study.

The correct answer is: All of the listed options

Question 7

Not answered

Marked out of 1

In a systematic review of studies looking at Duration of Untreated Psychosis and outcome, two types of study designs are noted. One group of studies categorically divided DUP to binary variable long DUP and short DUP and assessed the outcome. Another group of studies correlated the duration of DUP with the severity of the outcome. Which of the following best describes the above phenomenon?

Select one:

- ☐ Presence of location bias
- ☐ Presence of clinical heterogeneity
- ☐ Presence of methodological heterogeneity
- ☐ Presence of statistical heterogeneity
- ☐ Presence of publication bias

Your answer is incorrect.

Methodological heterogeneity refers to heterogeneity resulting from the differential use of study methodology. These may lead to different conclusions in different studies, even though the clinical characteristics are the same. Methodological heterogeneity is describable but does not need quantification.

The correct answer is: Presence of methodological heterogeneity

Question 8

Not answered

Marked out of 1

In the above-mentioned systematic review, which of the following is the best way to proceed for combining the data from individual studies?

Select one:

- ☐ Meta-analysis is not possible as the individual trials are not RCTs
- ☐ Combine all data in a single meta-analysis using a random effects model
- ☐ Combine all data and carry out a fixed effects analysis
- ☐ Deal with two sets of data differently and carry out two independent meta-analysis
- ☐ Meta-analysis must be abandoned as methods differ

Your answer is incorrect.

Meta-analysis is a best done using studies with a common methodology (all RCTs, all case-control studies) etc., But increasingly certain meta-analytical papers seem to override this tradition. Dealing with methodologically different data separately and reporting the results independently is the best way forward for the above example.

The correct answer is: Deal with two sets of data differently and carry out two independent meta-analysis

Question 9

Not answered

Marked out of 1

In which of the following situations a random effects analysis is indicated in a meta-analysis?

Select one:

- ☐ Presence of clinical heterogeneity
- ☐ Presence of too few studies for efficient combining
- ☐ Presence of statistical heterogeneity
- ☐ Meta-analysis of studies other than RCTs.
- ☐ Presence of methodological heterogeneity

Your answer is incorrect.

When combining the outcomes from different studies, one may use a fixed and/or random effects model. A fixed effects model assumes that all the studies share the same common treatment effect while a random effects model assumes that they do not share the same common treatment effect. In fixed effect analysis the inference is restricted to included set of studies. It assumes that only random error within studies could explain observed differences. It ignores between-study variations (hence heterogeneity). So this can be applied only if heterogeneity can be safely excluded by testing for it. Random effects analysis assumes that each study shows a different effect which are normally distributed around true mean. This assumption gives proportionally greater weight to smaller studies. Hence this model is susceptible to publication bias and results in wider less precise confidence intervals.

The correct answer is: Presence of statistical heterogeneity

Question 10

Not answered

Marked out of 1

A reviewer notes the following while undertaking a literature search to identify the effect of vitamin E on tardive dyskinesia. Many small studies are published only when they show a positive result in favour of vitamin E. But larger studies are published with both directions of results. If the reviewer constructs an inverted funnel plot which of the following will be seen?

Select one:

- ☐ The plot will be asymmetrical due to the absence of small negative studies.
- ☐ The plot cannot be constructed due to the lack of negative small studies
- ☐ The plot will be smaller in size due to the presence of small studies
- ☐ The plot will be asymmetrical due to the presence of large studies in both directions of results
- ☐ The plot will be symmetrical

Your answer is incorrect.

The most common method to detect publication bias is to investigate the presence of asymmetry in (inverted) funnel plots. A funnel plot shows the relation between study effect size and its precision. Small studies are more likely to remain unpublished if their results are nonsignificant or unfavourable, and larger studies get published regardless. This leads to funnel-plot asymmetry. Although visual inspection of funnel plots is unreliable, statistical tests can be used to quantify the asymmetry.

The correct answer is: The plot will be asymmetrical due to the absence of small negative studies.

Question 11

Not answered

Marked out of 1

Which of the following best describes effect size measured in meta-analyses?

Select one:

- ☐ Difference in outcome between intervention and control groups
- ☐ Ratio of outcome between intervention and control groups divided by standard deviation
- ☐ Difference in outcome between intervention and control groups in terms of standard deviation of the outcome in the population
- ☐ Odds of having an outcome due to intervention
- ☐ Ratio of outcome between intervention and control groups

Your answer is incorrect.

Effect size is the difference in mean scores between the treatment and control groups divided by the standard deviation of the scores in the control group (or the pooled standard deviation of the treatment and control groups, or the pooled standard deviation of the change score). As you might have noted, this does not have units that vary with outcome scales; hence it is possible to pool data from various studies in a meta-analysis using effect-size. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 736

The correct answer is: Difference in outcome between intervention and control groups in terms of standard deviation of the outcome in the population

Question 12

Not answered

Marked out of 1

Which one of the following could be classified as grey literature for the purpose of a systematic review?

Select one:

- ☐ Articles older than 30 years
- ☐ Peer reviewed articles indexed in PsycINFO
- ☐ Theses work recently submitted to universities
- ☐ Medline database
- ☐ Cochrane controlled trials register

Your answer is incorrect.

Searching for grey literature is often difficult as there is no single source of grey literature, and the sources are usually websites that are not indexed and can be difficult to search. Such grey literature includes theses, unpublished abstracts at conferences, monographs and proceedings from conventions, etc.

The correct answer is: Theses work recently submitted to universities

Question 13

Not answered

Marked out of 1

In a weekly journal club, your clinical tutor points out significant publication bias in the studies selected for a systematic review. He constructs a funnel plot to support his arguments. What is the most likely explanation for publication bias

Select one:

- ☐ Editors accept only those articles that are written by experts
- ☐ Systematic reviewers discard weaker studies in meta-analysis
- ☐ Researchers withhold negative results from publication
- ☐ Authors submit studies with low sample size to non-indexed journals
- ☐ Editors accept only those articles with which they agree

Your answer is incorrect.

Publication bias has been defined as "the tendency on the part of investigators to submit, or the reviewers and editors, to accept manuscripts based on the direction or strength of the study findings". As a result, the strongest and most positive studies are most likely to be published. "Perhaps the common misconception is that most work remains unpublished because of rejection from chosen journals. However, investigators have found the contrary to be true. Rejection of manuscripts by journals has been found to be less important than failure of researchers to actually submit their work for publication. Failure to submit manuscripts can be caused by a number of factors including unimportant results, incomplete analysis, and investigators being too busy or having lost interest. This lack of enthusiasm to publish has been coined the 'file drawer problem'. Dickersin K, Min Y, Meinert C. Factors influencing publication of research results. JAMA 1992; 267: 374-378 (Publication bias: raising awareness of a potential problem. Excerpts retrieved from <http://www.nature.com/bdj/journal/v194/n5/full/4809923a.html>).

The correct answer is: Researchers withhold negative results from publication

Question 14

Not answered

Marked out of 1

Meta-analyses that summarise data from multiple studies are needed because

Select one:

- ☐ Individual studies are susceptible to confounding
- ☐ Individual studies often use higher threshold of p values
- ☐ Individual studies are often biased
- ☐ Individual studies often lack sufficient power for generalised conclusions
- ☐ Individual studies are prone to measurement errors

Your answer is incorrect.

Remember GIGO principle in secondary research: Garbage in - Garbage out! The quality of a systematic review or meta-analysis depends on the quality of individual studies. Hence error or bias-prone individual studies cannot be replaced by a meta-analyses of those studies. But pooled data in a secondary study such as meta-analysis has larger sample size to achieve better precision.

The correct answer is: Individual studies often lack sufficient power for generalised conclusions

Question 15

Not answered

Marked out of 1

Who coined the term meta-analysis?

Select one:

- ☐ Cohen
- ☐ Shapiro
- ☐ Bayes
- ☐ Glass
- ☐ Kaplan

Your answer is incorrect.

Gene Glass coined the term in 1976 and described it as the "analysis of analyses".

The correct answer is: Glass



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